
Computer Networks
Lecture 10
Shmulik Kaplan

Today

- Recap
- Distance Vector (Bellman Ford)
  - Cost change
  - Poisoned reverse
  - Real life solution
- Routing Comparing & Summary
- Autonomous Systems
  - Defenition
  - Intra communication protocols
  - Inter communication protocol
- IP
  - Addressing introduction

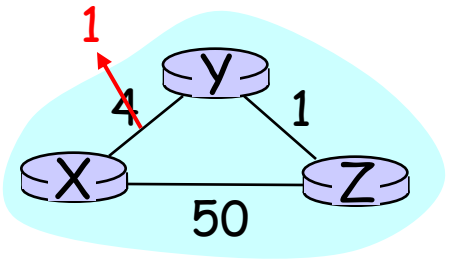
Today

- ✔ Recap
- Distance Vector (Bellman Ford)
  - Cost change
  - Poisoned reverse
  - Real life solution
- Routing Comparing & Summary
- Autonomous Systems
  - Definition
  - Intra communication protocols
  - Inter communication protocol
- IP
  - Addressing introduction

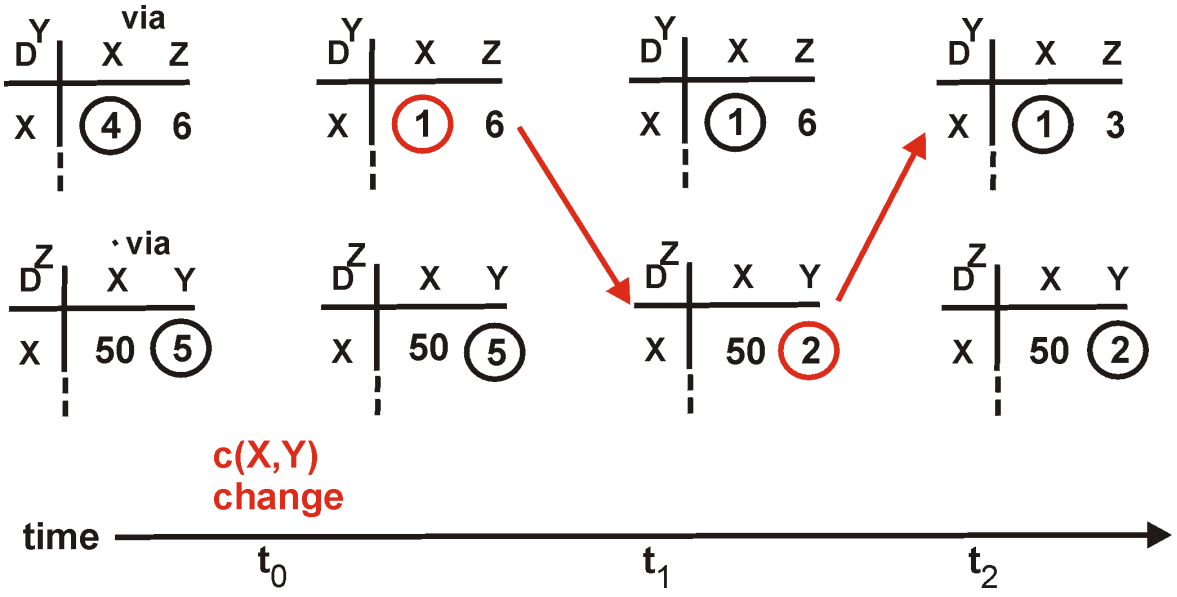
Distance Vector: link cost changes

Link cost changes:

- node detects local link cost change
- updates distance table (line 15)
- if cost change in least cost path, notify neighbors (lines 23,24)



"good news travels fast"

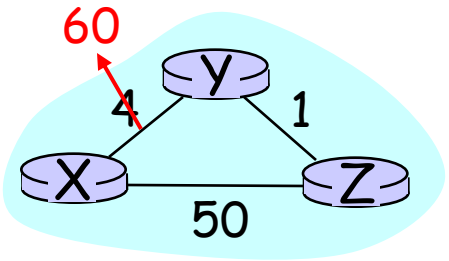


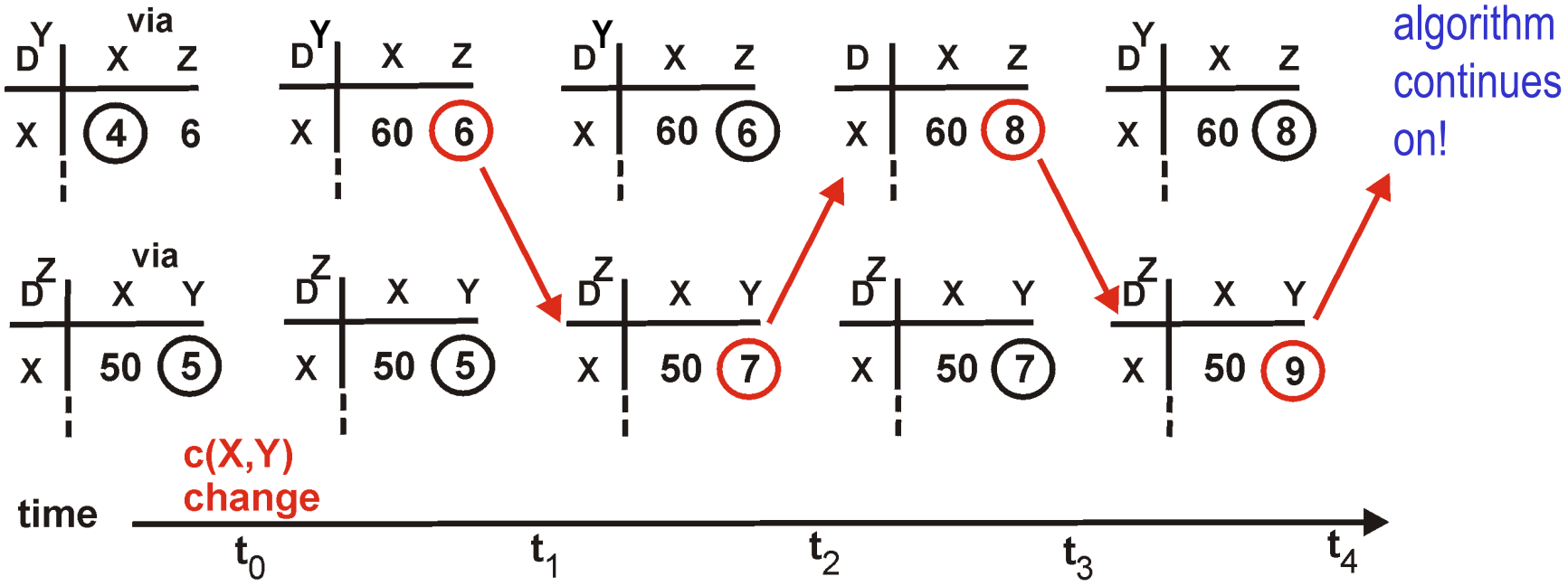
algorithm terminates

Distance Vector: link cost changes

Link cost changes:

- good news travels fast
- bad news travels slow – "count to infinity" problem!





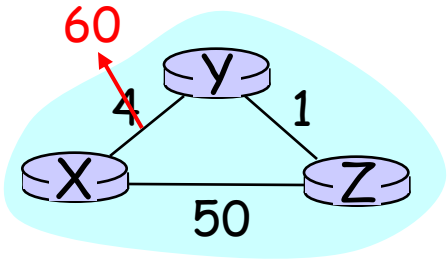
Today

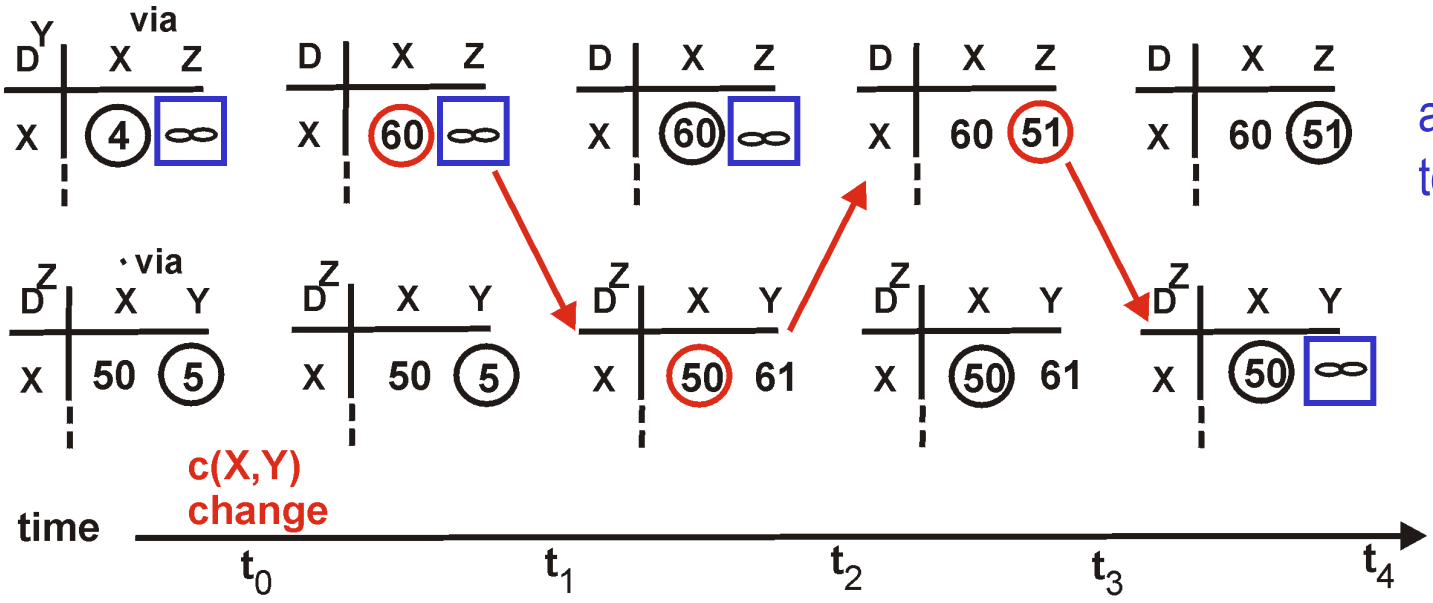
- ✓ Recap
- Distance Vector (Bellman Ford)
  - ✓ Cost change
  - Poisoned reverse
  - Real life solution
- Routing Comparing & Summary
- Autonomous Systems
  - Definition
  - Intra communication protocols
  - Inter communication protocol
- IP
  - Addressing introduction

Distance Vector: poisoned reverse

If Z routes through Y to get to X :

- Z tells Y its (Z's) distance to X is infinite (so Y won't route to X via Z)
- will this completely solve count to infinity problem?





Distance Vector Algorithm:

At all nodes, X:

1 Initialization:
2 for all adjacent nodes v:
3 $D^X(*,v) = \text{infty}$ /* the * operator means "for all rows" */
4 $D^X(v,v) = c(X,v)$
5 for all destinations, y
6 send $\min_w D^X(y,w)$ to each neighbor /* w over all X's neighbors */

Distance Vector Algorithm (cont.):

```
8 loop
9    wait (until I see a link cost change to neighbor V
10         or until I receive update from neighbor V)

12   if (c(X,V) changes by d)
13     /* change cost to all dest's via neighbor v by d */
14     /* note: d could be positive or negative */
15     for all destinations y:
16        D^X(y,V) = D^X(y,V) + d

17   else if (update received from V wrt destination Y)
18     /* shortest path from V to some Y has changed */
19     /* V has sent a new value for its min_W DV(Y,w) */
20     /* call this received new value is "newval" */
21     D^X(Y,V) = c(X,V) + newval

23   if we have a new min_W D^X(Y,w) in any destination Y
24     for each neighbor N:
25        for each destination Y:
26           if (nextHop(Y) == N)
27              send to N: D(Y) = ∞      /* poisoned reverse */
28           else
29              send to N: D(Y) = min_W D^X(Y,w)
30 forever
```

Today

- ✓ Recap
- Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
    - Real life solution
- Routing Comparing & Summary
- Autonomous Systems
  - Definition
  - Intra communication protocols
  - Inter communication protocol
- IP
  - Addressing introduction

Distance Vector: Real Life

Aim: reduce instability, not eliminate it entirely

- Poisoned reverse is not sufficient alone
- Protocols add multiple mechanisms:
  - Split horizon
  - Triggered updates
  - Hold-down timers
  - Bounded “infinity” values
- Cannot create a network that does both Dijkstra and Distance Vector

Today

- ✓ Recap
- ✓ Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
  - ✓ Real life solution
- Routing Comparing & Summary
- Autonomous Systems
  - Definition
  - Intra communication protocols
  - Inter communication protocol
- IP
  - Addressing introduction

Comparison of LS and DV algorithms

Message

- LS (Dijkstra): Initially O(nE) messages – each node has to know each link.
  Sent update messages to all nodes.
- DV: exchange between neighbors only (convergence time varies).

Robustness: **What happens if router malfunctions?**

- LS: node can advertise incorrect *link* cost,
  - each node computes only its *own* table
- DV: DV node can advertise incorrect *path* cost !!!
  - each node's table used by others (error propagate thru network)

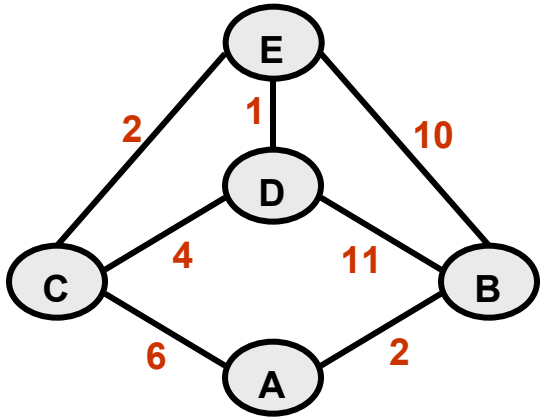
Distance Vector

A question from an exam

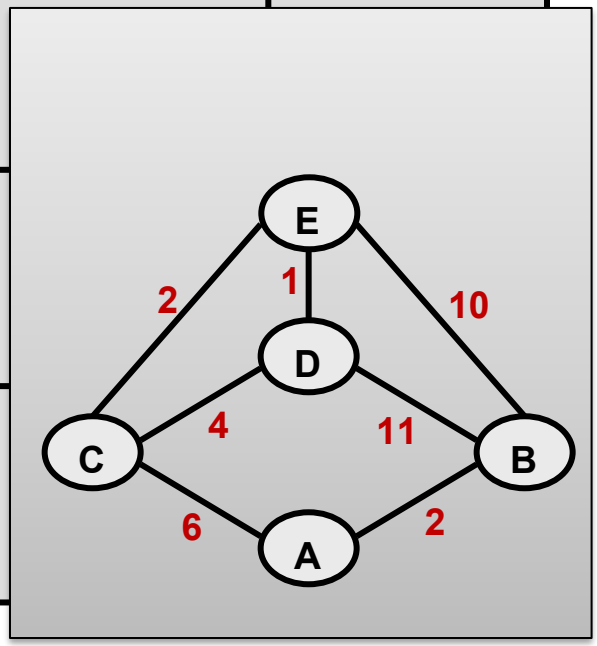
cost to destination via

destination

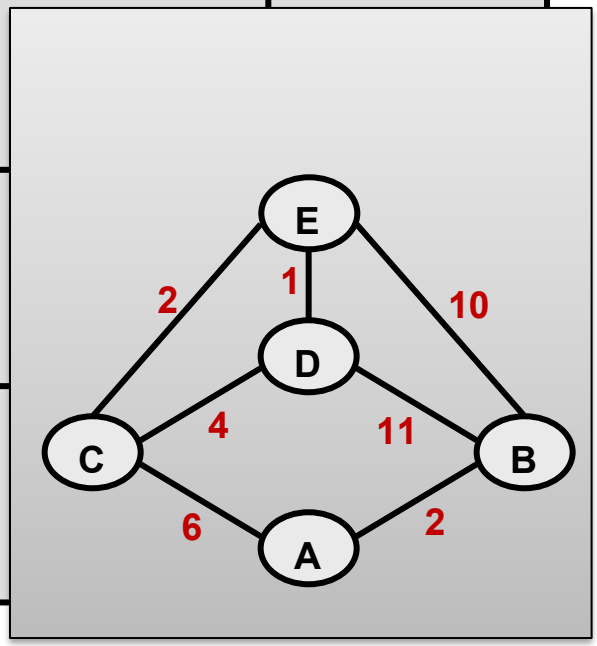
table of A	B	C
B	2	∞
C	∞	6
D	∞	∞
E	∞	∞



	t = 0	t = 1	t = 2	t = 3	t = 4	t = 5
	initial state	E→ D	D→ C	C→ D	D→ B	B→ A
A	B: B 2, C; C: 6; D; E					B; C; D; E
B	A; D; E; A; C; D; E				A; D; E; A; C; D; E	
C	A; D; E; A; B; D; E		A; D; E; A; B; D; E			
D	B; C; E; A; B; C; E	B; C; E; A; B; C; E		B; C; E; A; B; C; E		
E	B; C; D; A; B; C; D					



	t = 0	t = 1	t = 2	t = 3	t = 4	t = 5
	initial state	E→ D	D→ C	C→ D	D→ B	B→ A
A	B: B 2, C 6					B: B 2, C 6, D 13, E 12
B	A: A 2, D 11, E 10				A: A 2, D 21, C 14, D 11, E 12	
C	A: A 6, D 4, E 2		A: A 6, D 15, D 4, E 5			
D	B: B 11, C 4, E 1	B: B 11, C 4, E 11		B: B 11, C 10, C 19, C 4, E 6		
E	B: B 10, C 2, D 1					



Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

פתרו פעם אחת שלא קיימות בגרף קשתות מקבילות ופעם אחת כשקיימות קשתות מקבילות

אם הניתוב הוא לפי וקטור-מרחק, אזי כאשר יש שינוי כלשהו במשקל של אחת מהקשתות, תמיד צומת אחד לפחות חייב לשלוח הודעת עדכון.

1. נכון
2. לא נכון

In this question and the next ones, assume there can be parallel edges.

In distance-vector routing, when there is a change in the weight of one of the edges, at least one node must always send an update message.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

פתרו פעם אחת שלא קיימות בגרף קשתות מקבילות ופעם אחת כשקיימות קשתות מקבילות

אם הניתוב הוא לפי וקטור-מרחק, אזי כאשר יש שינוי כלשהו במשקל של אחת מהקשתות בעלות משקל מינימלי, תמיד צומת אחד לפחות חייב לשלוח הודעת עדכון.

1. נכון
2. לא נכון

In distance-vector routing, when there is any change in the weight of one of the minimal-weighted edges, at least one node must always send an update message.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

פתרו פעם אחת שלא קיימות בגרף קשתות מקבילות ופעם אחת כשקיימות קשתות מקבילות

אם הניתוב הוא לפי וקטור-מרחק, אזי כאשר יש שינוי כלשהו במשקל של אחת מהקשתות ולאחר השינוי קשת זו היא אחת מהקשתות בעלות המשקל המינימלי, תמיד צומת אחד לפחות חייב לשלוח הודעת עדכון.

1. נכון
2. לא נכון

In distance-vector routing, when there is a change in the weight of one of the minimal-weighted edges, and its weight decreases, at least one node must always send an update message.

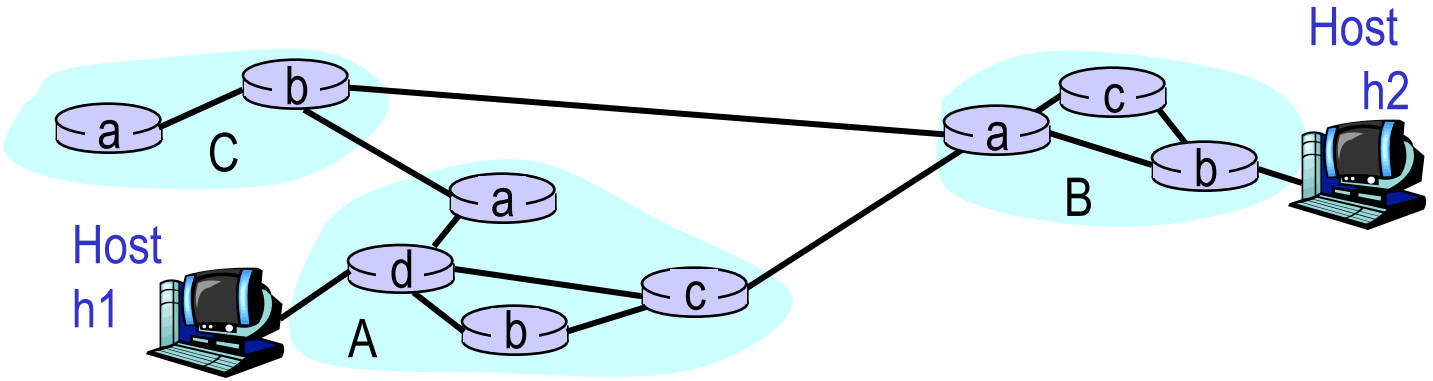
1. True
2. False

Today

- ✓ Recap
- ✓ Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
  - ✓ Real life solution
- ✓ Routing Comparing & Summary
- Autonomous Systems
  - Definition
  - Intra communication protocols
  - Inter communication protocol
- IP
  - Addressing introduction

Intra-AS and Inter-AS routing

(AS = Autonomous System)

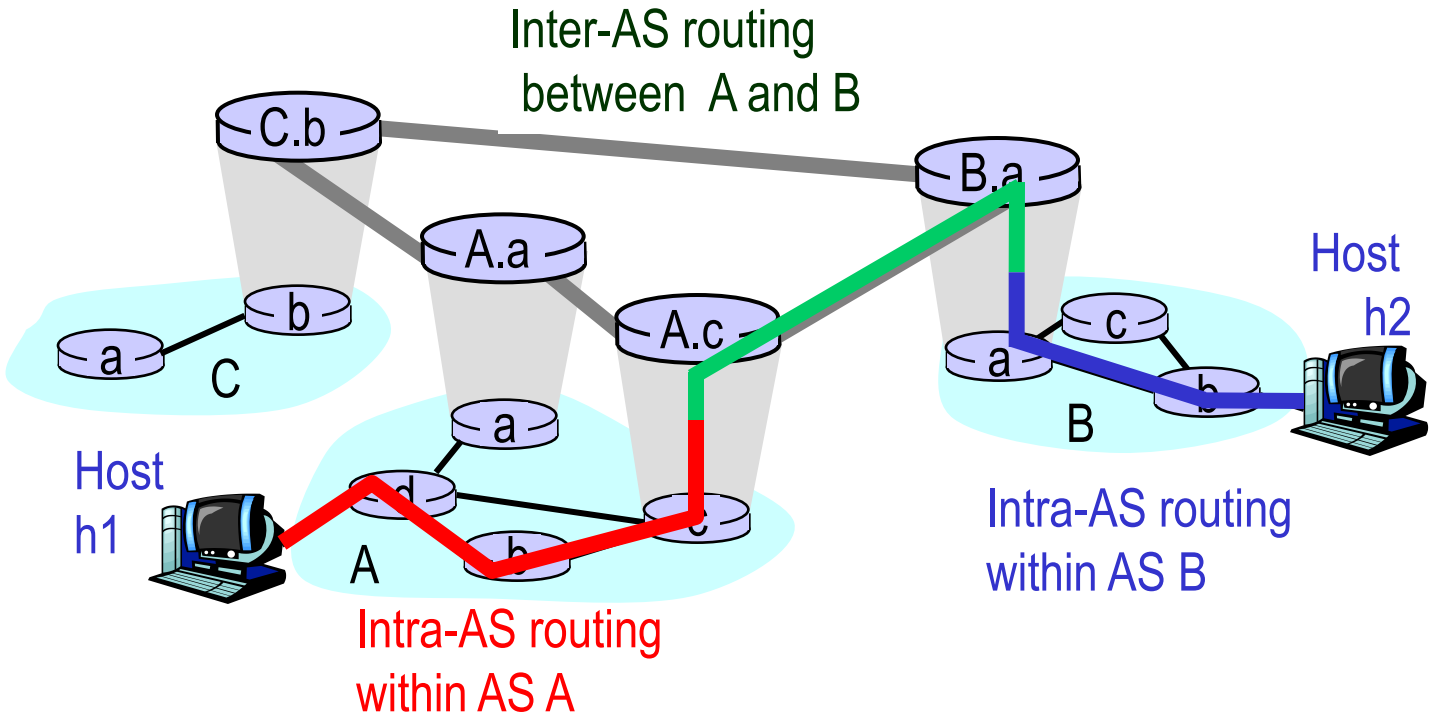


Two-level routing:

- **Intra-AS:** administrator is responsible for choice
- **Inter-AS:** unique standard

Intra-AS and Inter-AS routing

(AS = Autonomous System)



Two-level routing:

- **Intra-AS:** administrator is responsible for choice
- **Inter-AS:** unique standard

Today

- ✓ Recap
- ✓ Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
  - ✓ Real life solution
- ✓ Routing Comparing & Summary
- Autonomous Systems
  - ✓ Definition
    - Intra communication protocols
    - Inter communication protocol
- IP
    - Addressing introduction

Routing the internet: Intra-AS Routing

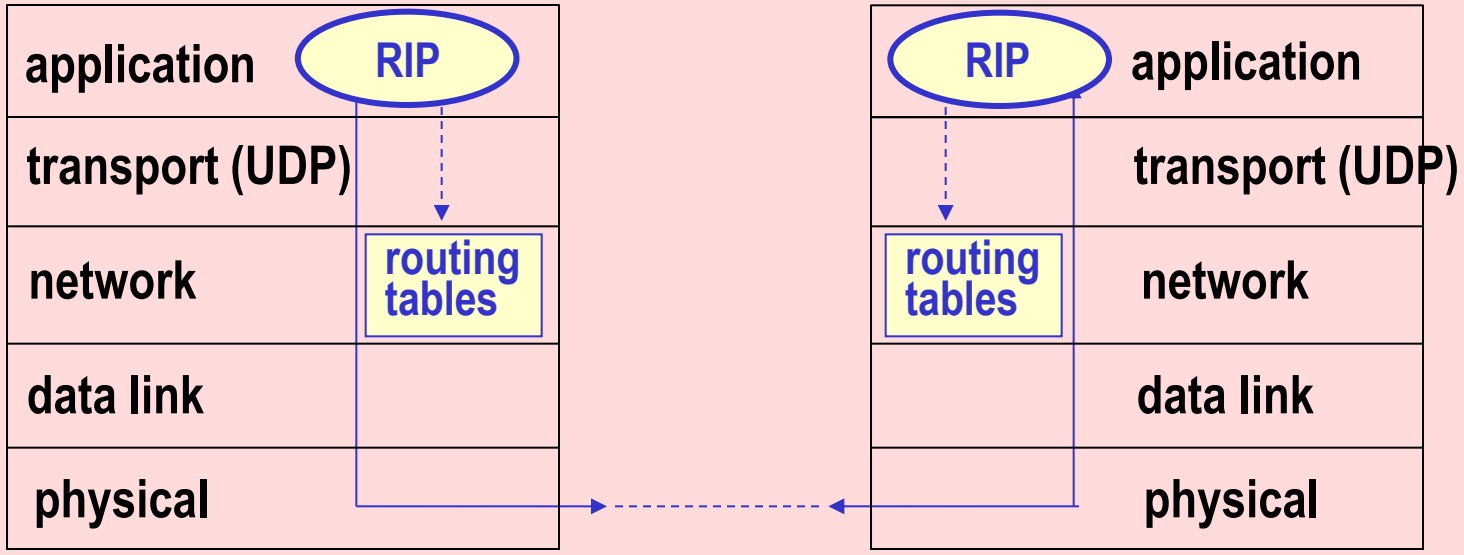
- Also known as Interior Gateway Protocols (IGP)
- Most common IGPs:
  - RIP: Routing Information Protocol
  - IGRP: Interior Gateway Routing Protocol (Cisco propr.)
  - OSPF: Open Shortest Path First
  - IS-IS: Intermediate System to Intermediate System

RIP (Routing Information Protocol)

- Distance vector algorithm
- Included in BSD-UNIX Distribution in 1982
- Each link has cost 1
- The max cost of a path is limited to 15.
- Updates are exchanged every 30 sec via *response message* (also called **advertisement**)
- If no advertisement heard after 180 sec ➔ neighbor/link declared dead
- poison reverse used to prevent ping-pong loops (infinite distance = 16 hops)

RIP Table processing

- RIP routing tables managed by application-level process
- advertisements sent in UDP packets, periodically repeated



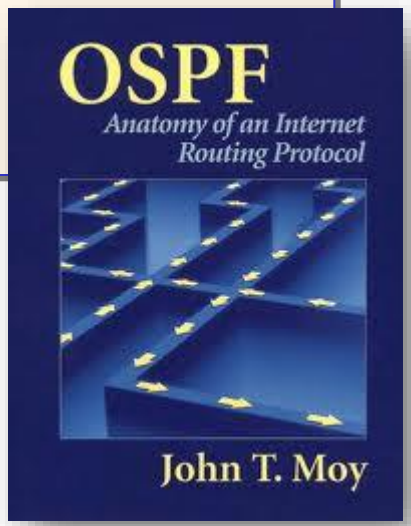
IGRP (Interior Gateway Routing Protocol)

- CISCO proprietary; successor of RIP (mid 80s)
- Distance Vector, like RIP
- several cost metrics (delay, bandwidth, reliability, load etc)
- uses TCP to exchange routing updates
- Properties:
  - Good algorithm quality
  - Converges quickly
  - Handles loops well
  - Scalable

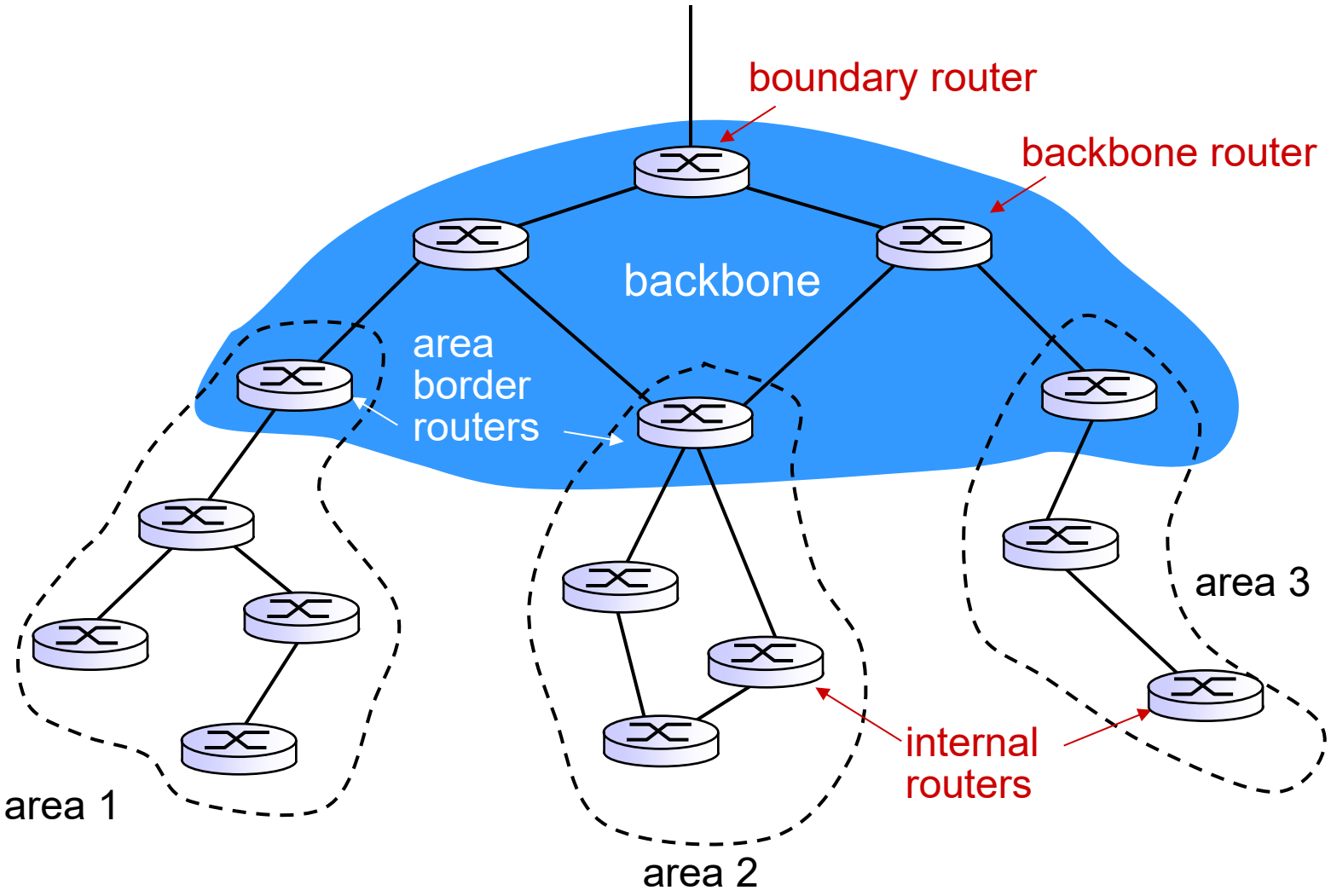
OSPF (Open Shortest Path First)

- "open": publicly available
- Uses link-state (Dijkstra's) algorithm
- Advertisements disseminated to entire AS (via flooding)
- Carried in OSPF messages directly over IP (rather than TCP or UDP)

- Security: all OSPF messages authenticated (to prevent malicious intrusion)
- Hierarchical OSPF in large domains



Hierarchical OSPF

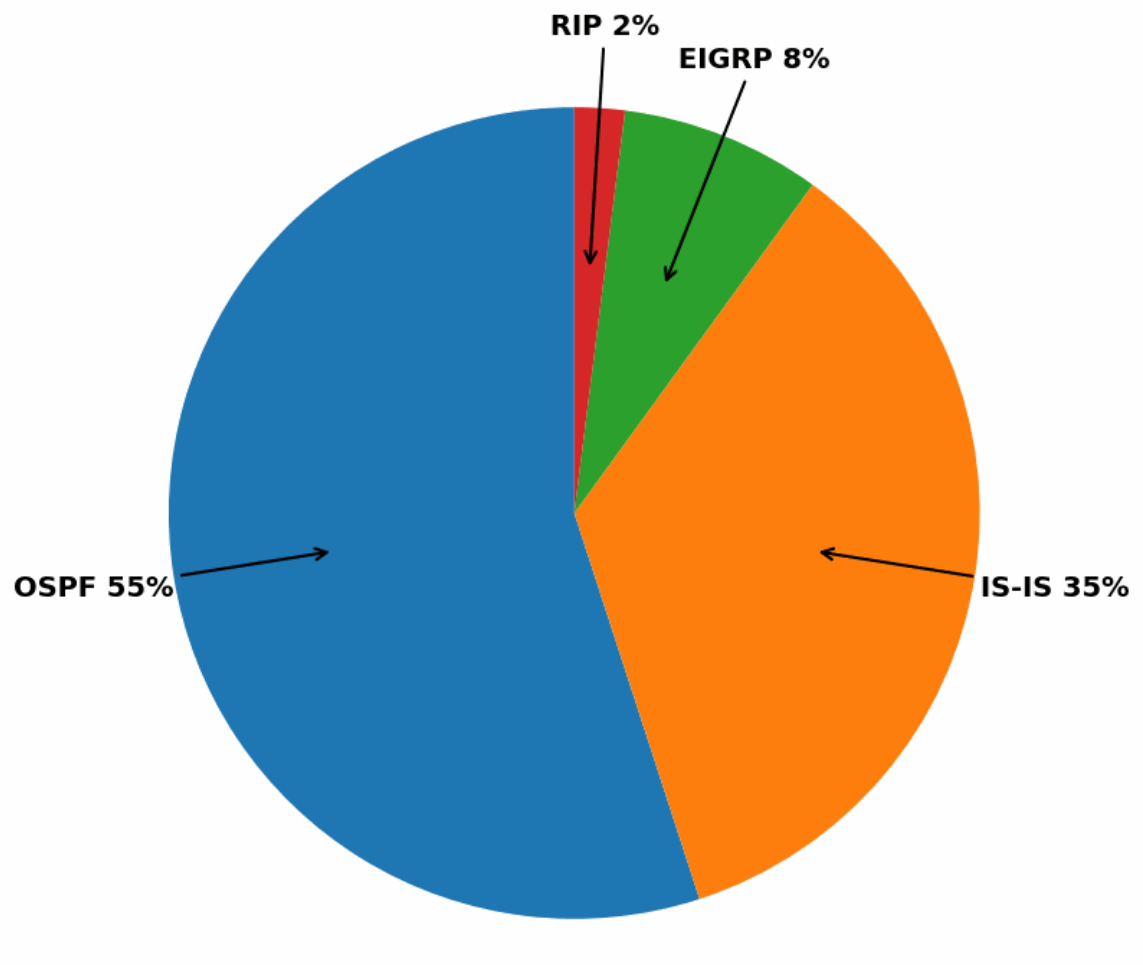


IS-IS (Intermediate System to Intermediate System

- Link-state routing protocol (uses Dijkstra's algorithm)
- Originally defined by ISO for CLNP, later extended for IP (Integrated IS-IS)
- Advertisements (LSPs) flooded throughout the AS (similar to OSPF)
- Runs directly over Layer-2 (no TCP/UDP/IP transport)

- Hierarchical design:
  - Level-1: Intra-area routing
  - Level-2: Inter-area routing
- Highly scalable and stable widely used in ISP backbones
- Supports multiple metrics (default: cost)
- Fast convergence; minimal protocol overhead

AS – protocol current usability



Why Cisco lost most of the market share

- Originally was Proprietary:
  Operators prefer NOT to be dependent on a single vendor
- Opened too late:
  Operators rarely replace existing solutions with no really good reason
- Industry prefers standards
  Open standards are very well viewed and debugged
  Operators like the transparency
- Not adopted by the big ISPs and Internet core
  Big ISPs chose IS-IS and OSPF
- Operational and educational momentum
  Being open induced a “community” from universities and other courses – many people with knowhow
- Human perception
  Once considered Proprietary – people still think of it as such even if it was opened

Question from an exam
Circle the correct answer

שאלה מבחינה
הקף בעיגול את התשובה המתאימה

השאלה מתייחסת לפרוטוקולים RIP ו- OSPF.

1. שניהם רצים מעל אותו פרוטוקול ברמת התובלה.
2. שניהם מבוססים על אותו אלגוריתם ניתוב (distance vector או link-state).
3. בשניהם עדכון במשקל של קשת מופץ לכל הצמתים ברשת.
4. כל התשובות (למעלה) נכונות.
5. כל התשובות (למעלה) לא נכונות.

The question refer to the protocols RIP and OSPF

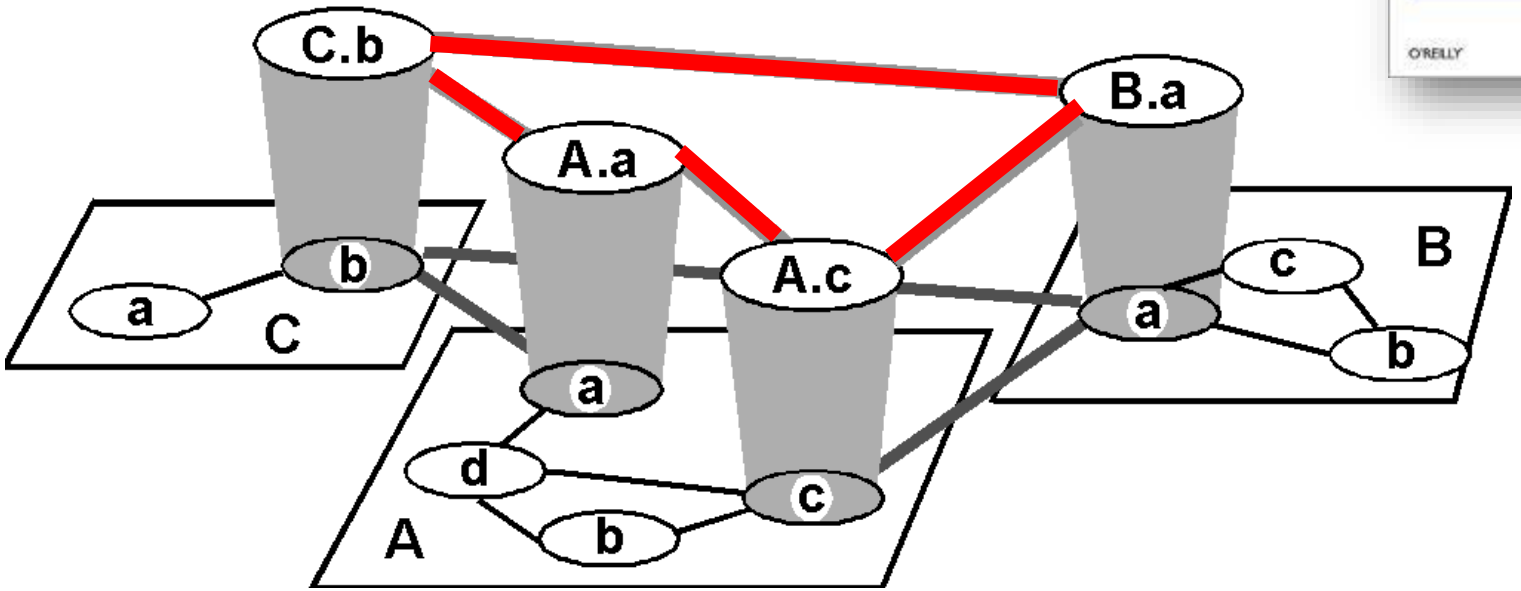
1. Both run over the same protocol in the transport layer.
2. Both are based on the same routing algorithm (link-state or distance vector).
3. In both a change in the weight of a link is distributed to all the nodes in the net.
4. All of the above.
5. None of the above.

Today

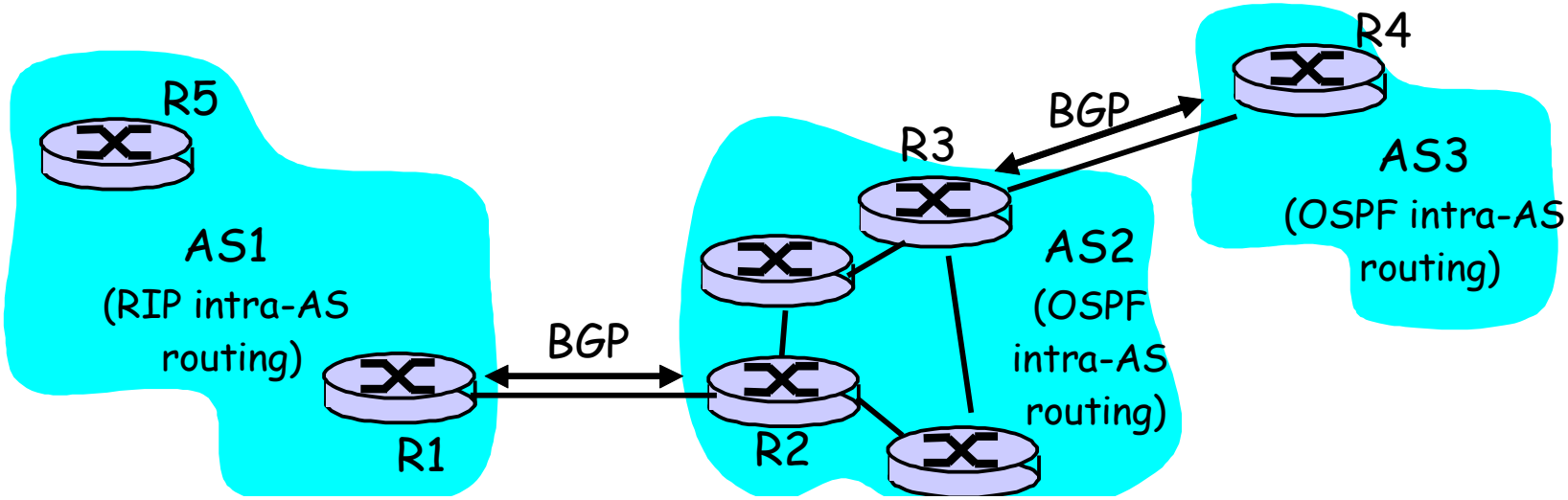
- ✓ Recap
- ✓ Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
  - ✓ Real life solution
- ✓ Routing Comparing & Summary
- Autonomous Systems
  - ✓ Definition
  - ✓ Intra communication protocols
    - Inter communication protocol
- IP
    - Addressing introduction

Inter-AS routing





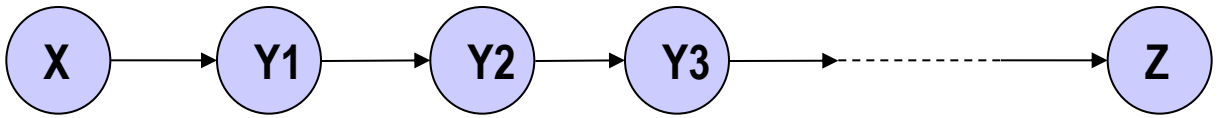
Inter-AS routing in the Internet: BGP



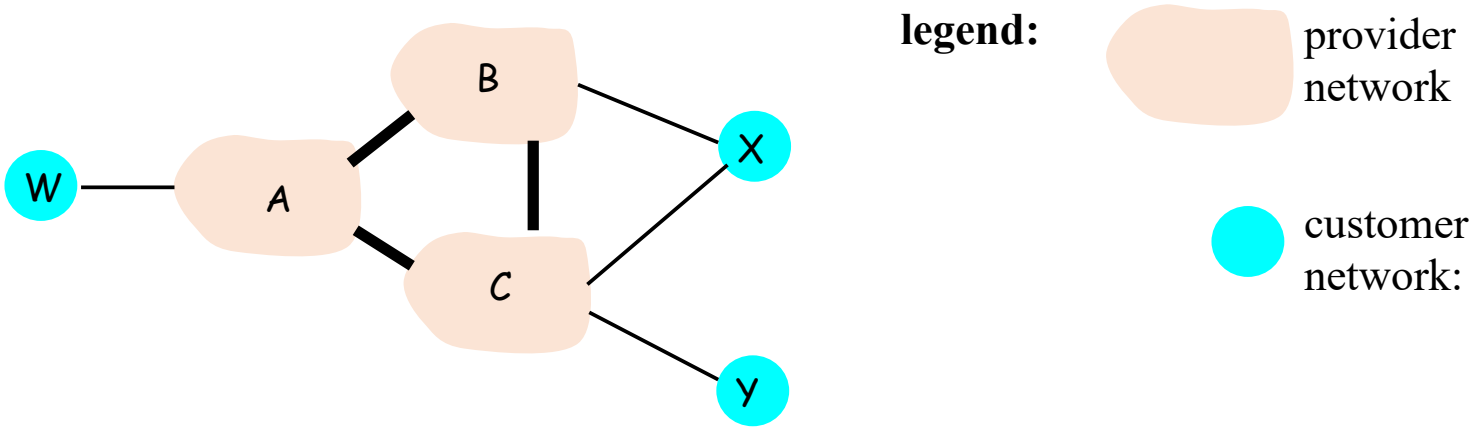
Internet inter-AS routing: BGP

- BGP (Border Gateway Protocol): *the* de facto standard
- Path Vector protocol
  - Similar to Distance Vector protocol
  - Each Border Gateway broadcast to neighbors (peers) *entire path* (I.e, sequence of ASs) to destination
  - E.g., Gateway X may send its path to dest. Z:

Path (X,Z) = X,Y1,Y2,Y3,…,Z

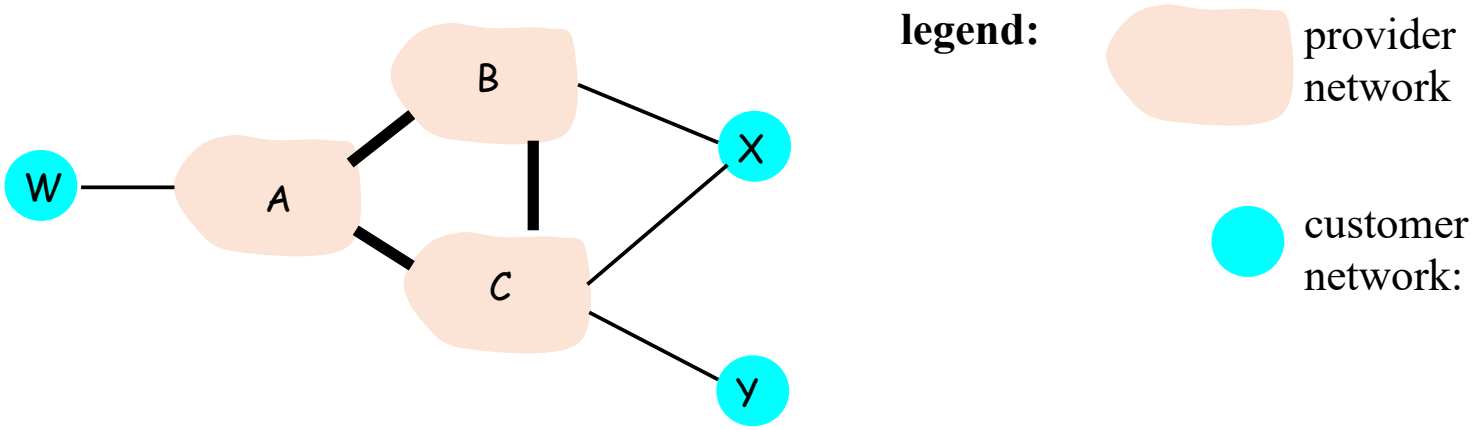


BGP: controlling who routes to you



- A,B,C are provider networks
- X,W,Y are customer (of provider networks)
- X is dual-homed: attached to two networks
  - X does not want to route from B via X to C
  - .. so X will not advertise to B a route to C

BGP: controlling who routes to you



- A advertises to B the path AW
- B advertises to x the path BAW
- Should B advertise to C the path BAW?
  - No way! B gets no "revenue" for routing CBAW since neither W nor C are B's customers
  - B wants to force C to route to w via A
  - B wants to route *only* to/from its customers!

BGP operation



BGP messages

- BGP messages exchanged over TCP.
- BGP messages:
  - OPEN: opens TCP connection to peer and authenticates sender
  - UPDATE: advertises new path (or withdraws old)
  - KEEPALIVE: keeps connection alive in absence of UPDATES; also ACKs OPEN request
  - NOTIFICATION: reports errors in previous msg; also used to close connection

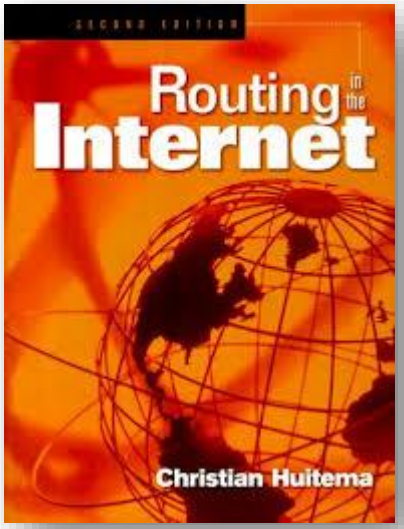
Why different Intra- and Inter-AS routing ?

- Policy
  - Inter-AS: admin wants control over how its traffic routed, who routes through its net
  - Intra-AS: single admin, so no policy decisions needed
- Scale
  - hierarchical routing saves table size, reduced update traffic
- Performance
  - Intra-AS: can focus on performance
  - Inter-AS: policy may dominate over performance

Want to read more about routing...

Routing in the Internet, 2nd Edition

By christian Huitema

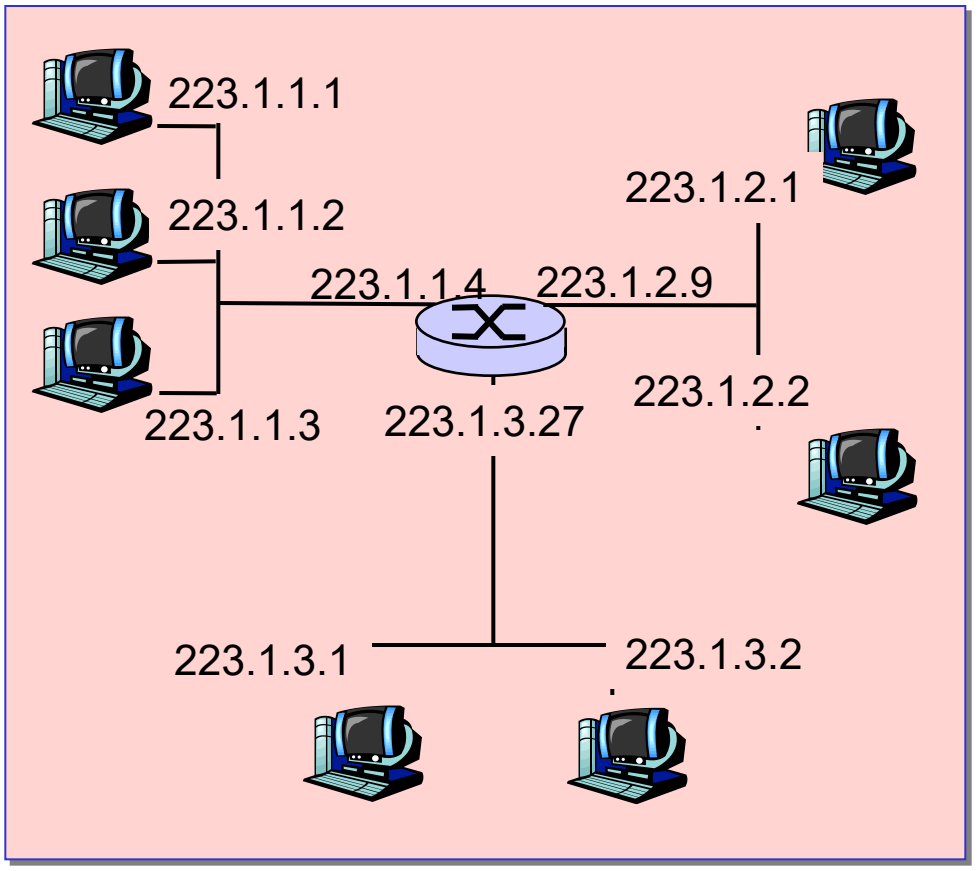


Today

- ✓ Recap
- ✓ Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
  - ✓ Real life solution
- ✓ Routing Comparing & Summary
- ✓ Autonomous Systems
  - ✓ Definition
  - ✓ Intra communication protocols
  - ✓ Inter communication protocol
- IP
  - Addressing introduction

IP Addressing: introduction

- IP address: 32-bit identifier for host, router *interface*
- *interface:* connection between host, router and physical link
  - router's typically have multiple interfaces
  - host may have multiple interfaces
  - IP addresses associated with interface, not host, router

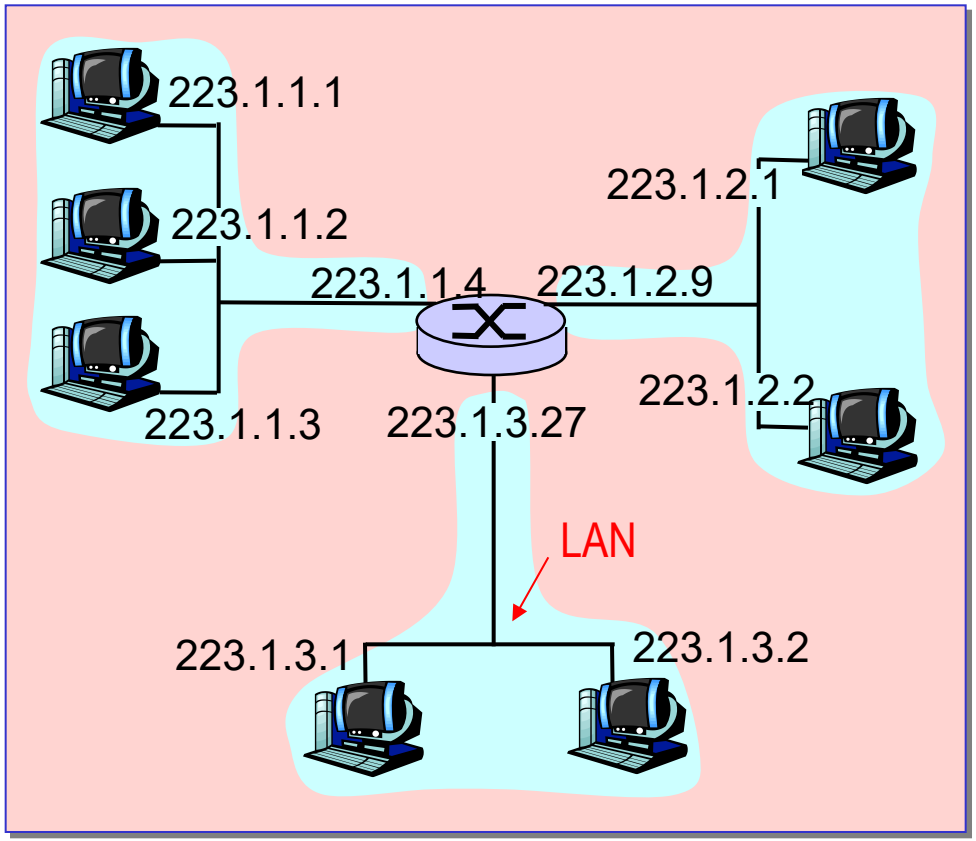


223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

IP Addressing

network part (high order bits) + host part (low order bits)

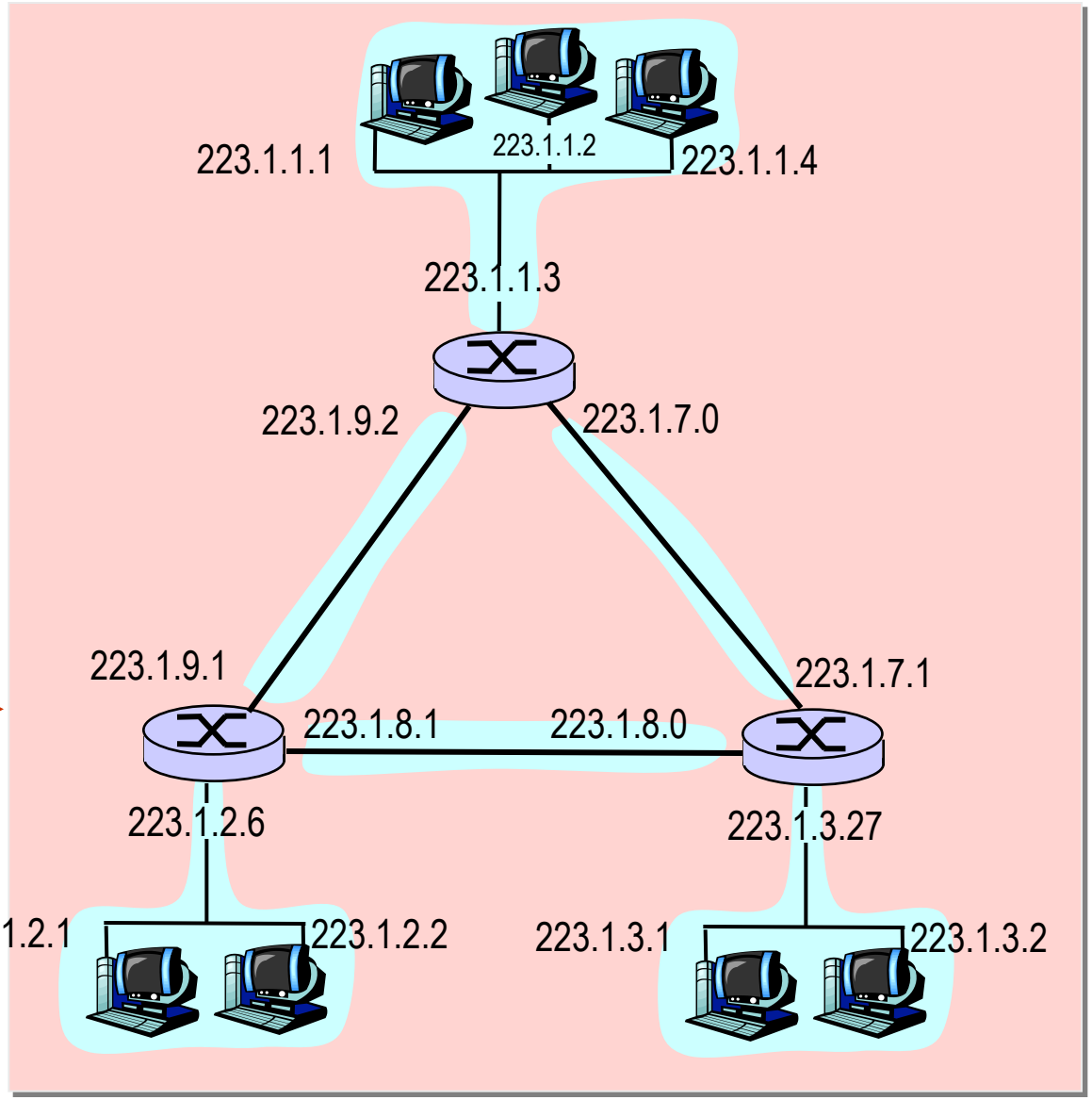


IP Addressing

How to find the networks?

- Detach each interface from router, host
- create “islands of isolated networks”

Interconnected system consisting of six networks



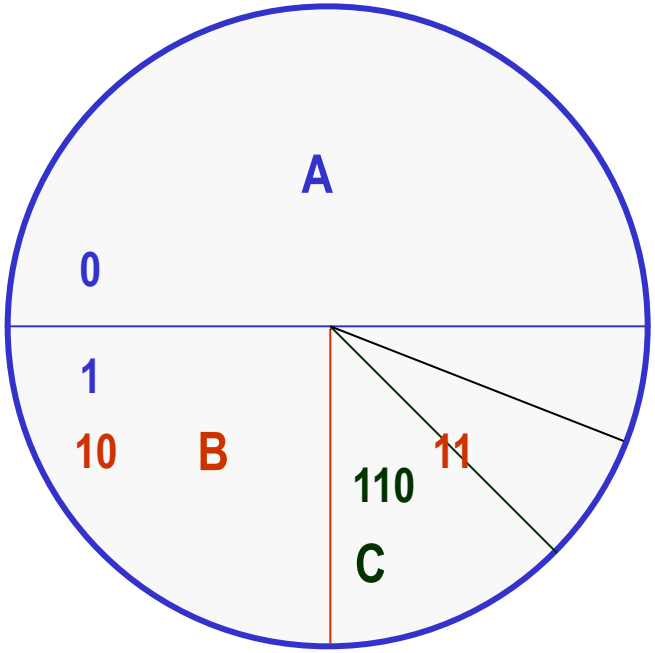
IP Addresses

"class-full" addressing

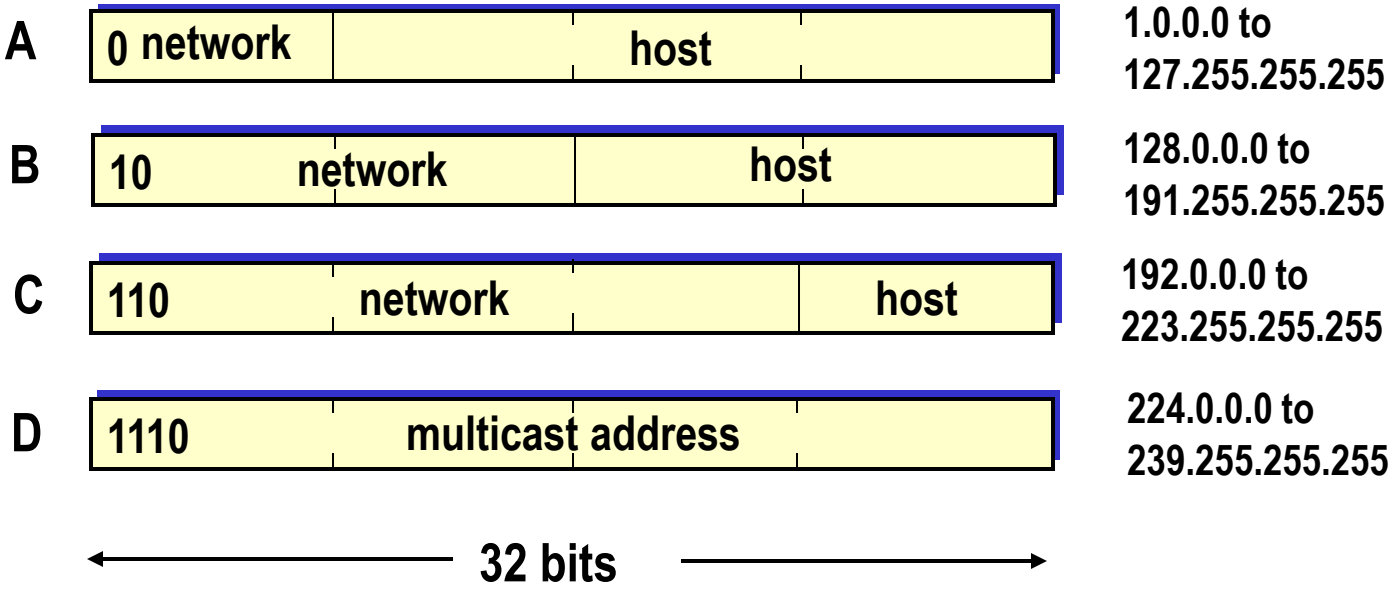
A B C

of networks:

Size of network:

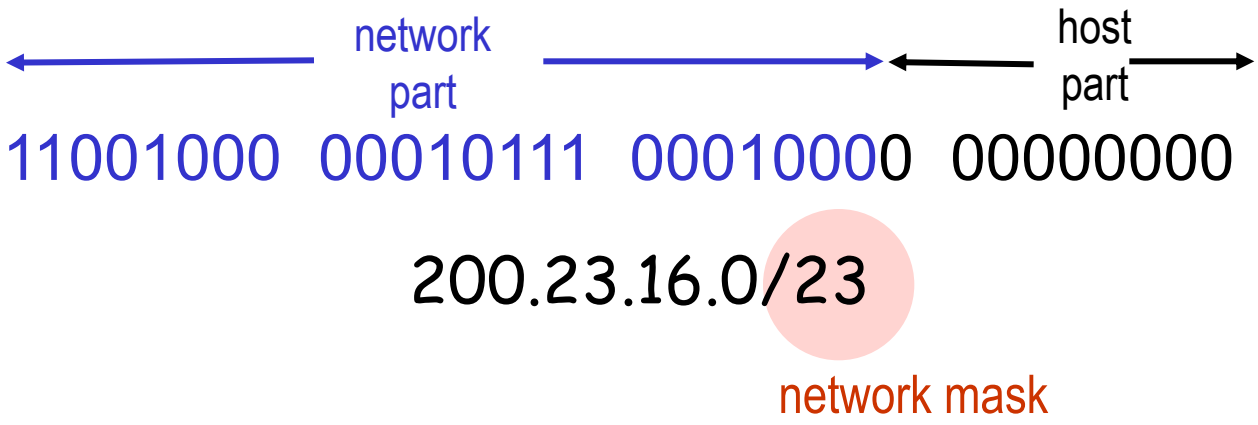


class



IP addressing: CIDR

- Classful addressing:
  - inefficient use of address space, address space exhaustion
  - e.g., class B net allocated enough addresses for 65K hosts, even if only 2K hosts in that network
- CIDR: Classless InterDomain Routing
  - network portion of address of arbitrary length
  - address format: a.b.c.d/x, where x is # bits in network portion of address



IP addresses: how to get one?

Hosts (host portion):

- hard-coded by system admin in a file
- DHCP: Dynamic Host Configuration Protocol: dynamically get address:
  - host broadcasts “DHCP discover” msg
  - DHCP server responds with “DHCP offer” msg
  - host requests IP address: “DHCP request” msg
  - DHCP server sends address: “DHCP ack” msg

(more shortly)

IP addresses: how to get one?

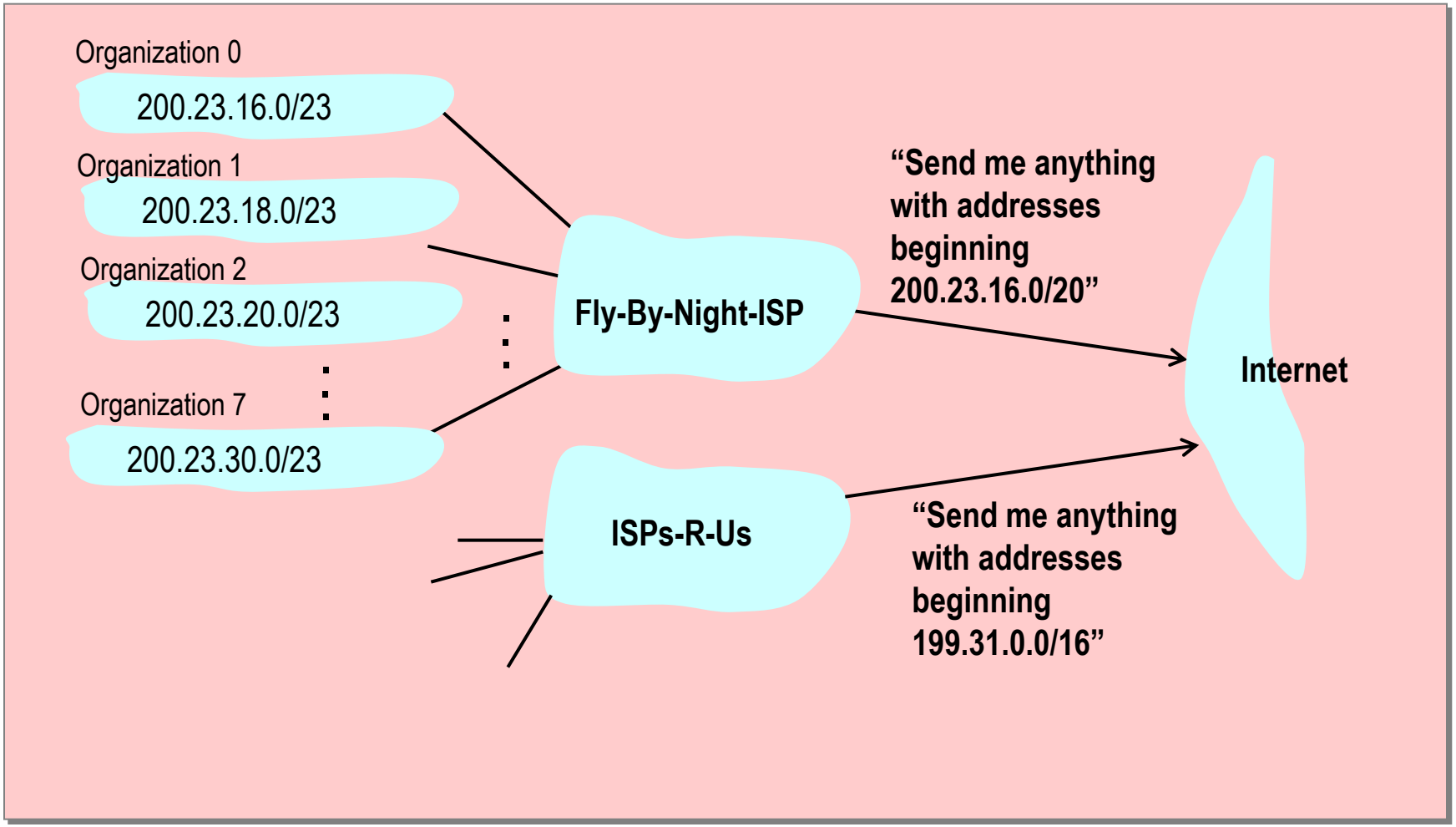
Network (network portion):

- get allocated portion of ISP's address space:

ISP's block	11001000 00010111 00010000 00000000	200.23.16.0/20
Organization 0	11001000 00010111 00010000 00000000	200.23.16.0/23
Organization 1	11001000 00010111 00010010 00000000	200.23.18.0/23
Organization 2	11001000 00010111 00010100 00000000	200.23.20.0/23
...	.....	....
Organization 7	11001000 00010111 00011110 00000000	200.23.30.0/23

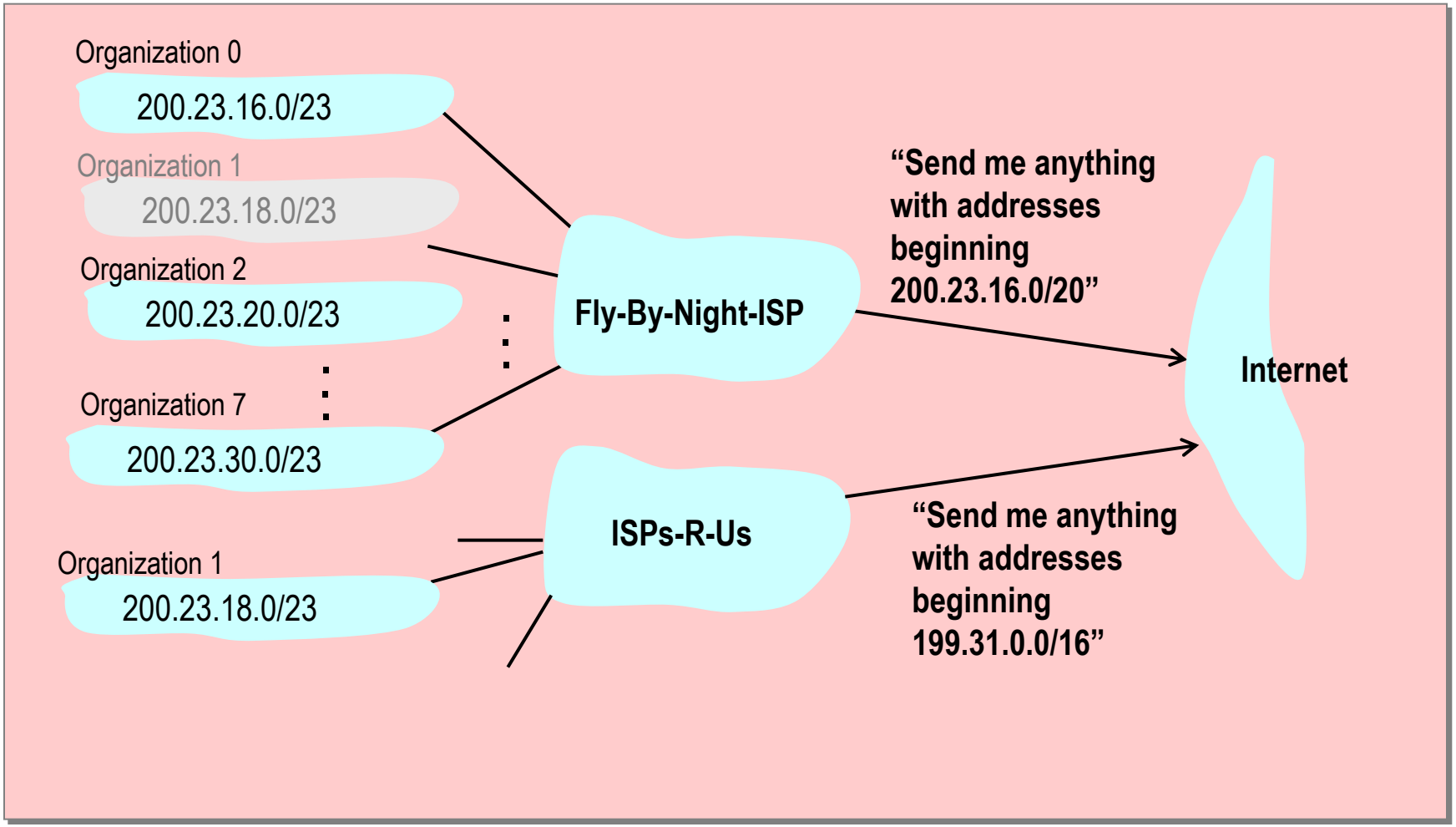
Hierarchical addressing: route aggregation

Hierarchical addressing allows efficient advertisement of routing information



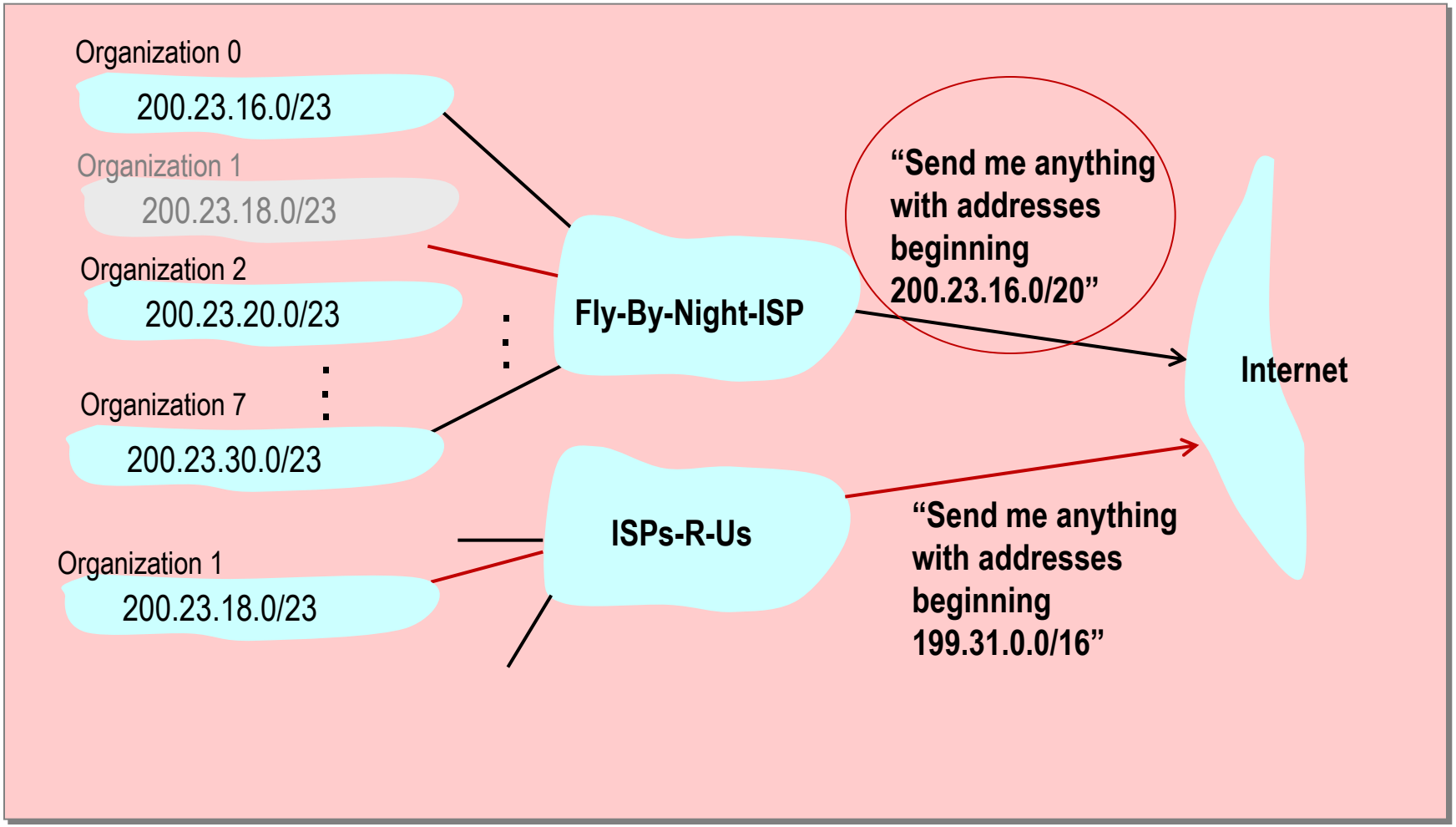
Hierarchical addressing: route aggregation

Hierarchical addressing allows efficient advertisement of routing information



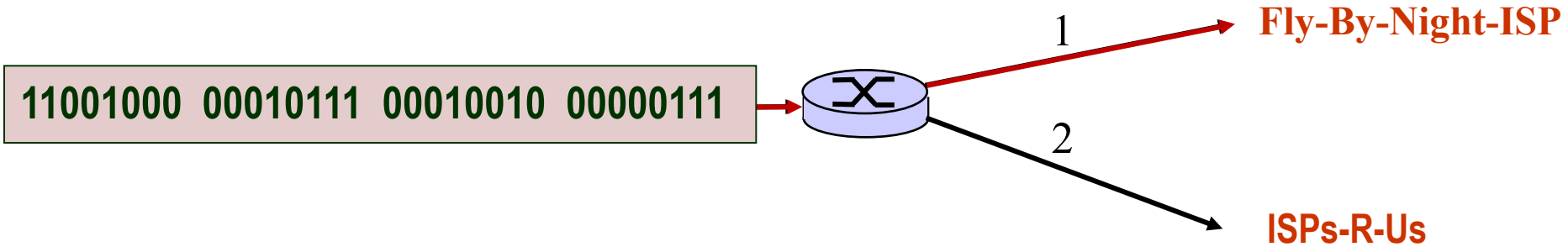
Hierarchical addressing: route aggregation

Hierarchical addressing allows efficient advertisement of routing information



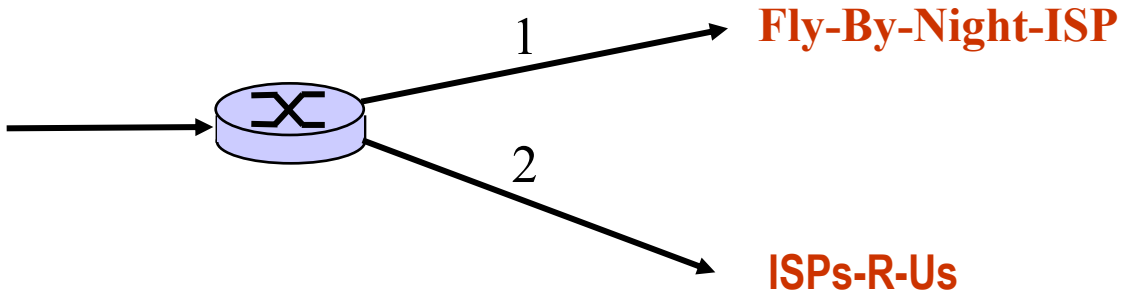
Longest prefix – before migration

Fly-By-Night-ISP	11001000 00010111 00010000 00000000 200.23.16.0/20	1
ISPs-R-Us	11000111 00011111 00000000 00000000 199.31.0.0/16	2



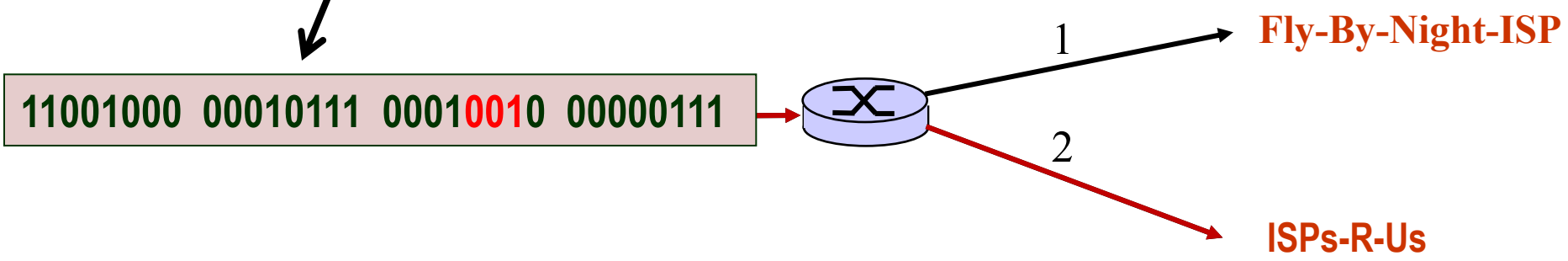
Longest prefix - changes

Fly-By-Night-ISP	11001000 00010111 00010000 00000000 200.23.16.0/20	1
ISPs-R-Us	11000111 00011111 00000000 00000000 199.31.0.0/16	2



Longest prefix - works

Fly-By-Night-ISP	11001000 00010111 00010000 00000000 200.23.16.0/20	1
ISPs-R-Us (Organization 1)	11001000 00010111 00010010 00000000 200.23.18.0/23	2
ISPs-R-Us	11000111 00011111 00000000 00000000 199.31.0.0/16	2



Today

- ✓ Recap
- ✓ Distance Vector (Bellman Ford)
  - ✓ Cost change
  - ✓ Poisoned reverse
  - ✓ Real life solution
- ✓ Routing Comparing & Summary
- ✓ Autonomous Systems
  - ✓ Definition
  - ✓ Intra communication protocols
  - ✓ Inter communication protocol
- ✓ IP
  - ✓ Addressing introduction

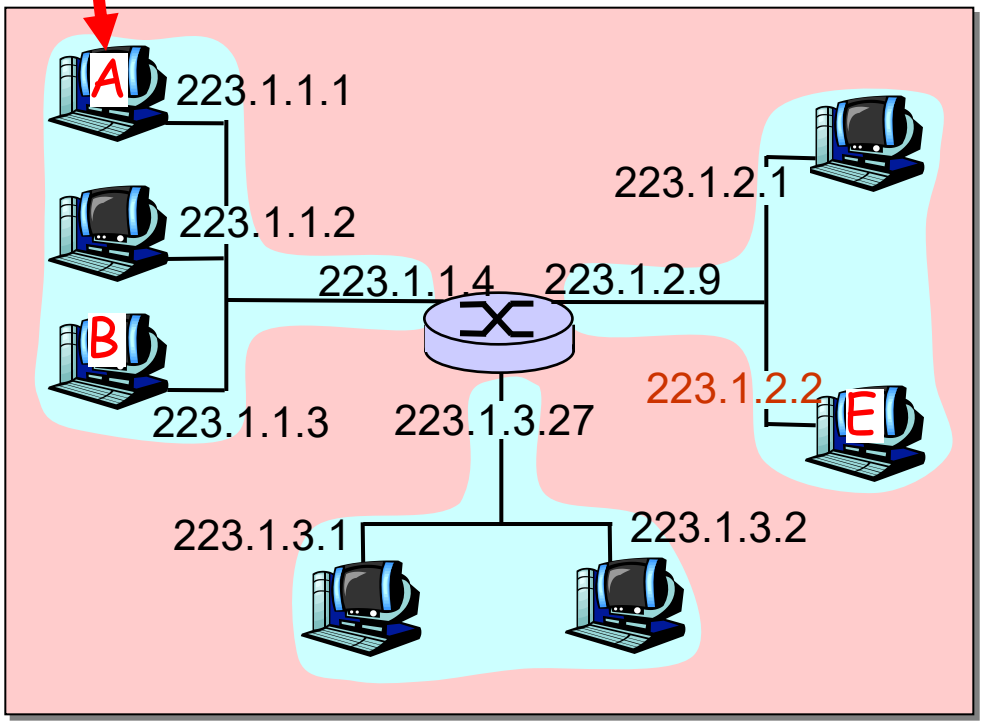
Getting a datagram from source to dest.

misc fields	223.1.1.1	223.1.2.2	data

Starting at A, dest. E

- look up network address of E
- E on *different* network
  - A, E not directly attached
- routing table: next hop router to E is 223.1.1.4
- link layer sends datagram to router 223.1.1.4 inside link-layer frame
- datagram arrives at 223.1.1.4
- continued…..

Dest. Net.	next router	Nhops
223.1.1		1
223.1.2	223.1.1.4	2
223.1.3	223.1.1.4	2



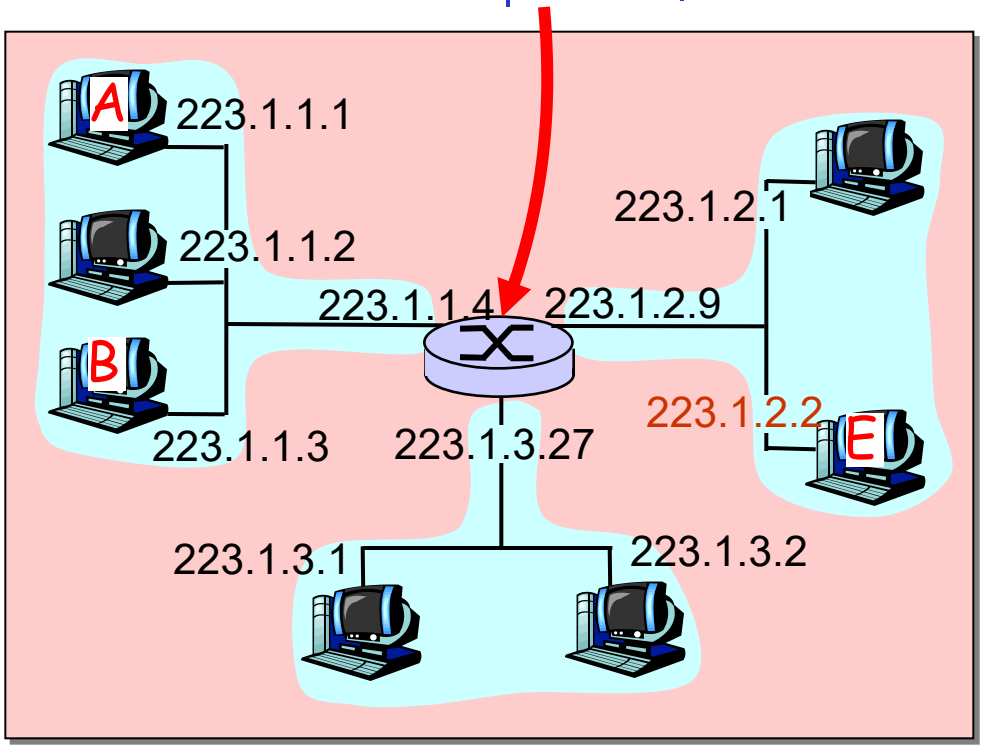
Getting a datagram from source to dest.

misc fields	223.1.1.1	223.1.2.2	data

Dest. network	next router	Nhops	interface
223.1.1	-	1	223.1.1.4
223.1.2	-	1	223.1.2.9
223.1.3	-	1	223.1.3.27

Arriving at 223.1.4, destined for 223.1.2.2

- look up network address of E
- E on *same* network as router's interface 223.1.2.9
  - router, E directly attached
- link layer sends datagram to 223.1.2.2 inside link-layer frame via interface 223.1.2.9
- datagram arrives at 223.1.2.2!!!



Example: IP addressing and routing

- Millions of addresses are available starting at 194.24.0.0
- Cambridge U. asks for 2048 addresses, then
- Oxford U. asks for 4096 addresses, then
- Edinburgh U. asks for 1024 addresses.

- How to assign the addresses? You have to assign addresses immediately as the request arrives.

Example: IP addressing and routing

Starting at 194.24.0.0; Cambridge 2048; Oxford 4096; Edinburgh 1024.

University	First address	Last address	How many	Mask
Cambridge	194.24.0.0	194.24.7.255	2048	194.24.0.0/21
Edinburgh	194.24.8.0	194.24.11.255	1024	194.24.8.0/22
(Available)	194.24.12.0	194.24.15.255	1024	194.24.12.0/22
Oxford	194.24.16.0	194.24.31.255	4096	194.24.16.0/20

Address	Mask
C: 11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000
C: 11000010 00011000 00000111 11111111	
E: 11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000
E: 11000010 00011000 00001011 11111111	
O: 11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000
O: 11000010 00011000 00011111 11111111	

Example: IP addressing and routing

Address	Mask	Link
C: 11000010 00011000 00000000 00000000	11111111 11111111 11111000 00000000	0
E: 11000010 00011000 00001000 00000000	11111111 11111111 11111100 00000000	3
O: 11000010 00011000 00010000 00000000	11111111 11111111 11110000 00000000	3

A packet comes in addressed to 194.24.17.4 , in binary:

11000010 00011000 00010001 00000100

First, it is ANDed with Cambridge mask to get

11000010 00011000 00010000 00000000 does not match

Next, it is ANDed with Edinburgh mask to get

11000010 00011000 00010000 00000000 does not match

Next, it is ANDed with Oxford mask to get

11000010 00011000 00010000 00000000 match

Aggregate entry 194.24.0.0/19

11000010 00011000 00000000 00000000 11111111 11111111 11100000 00000000

Longest prefix matching

Prefix Match	Link Interface
11001000 00010111 00010	0
11001000 00010111 00011	1
11001000 00010111 00011000	2
otherwise	3

Examples

DA: 11001000 00010111 00010110 10100001 Which interface?

DA: 11001000 00010111 00011000 10101010 Which interface?

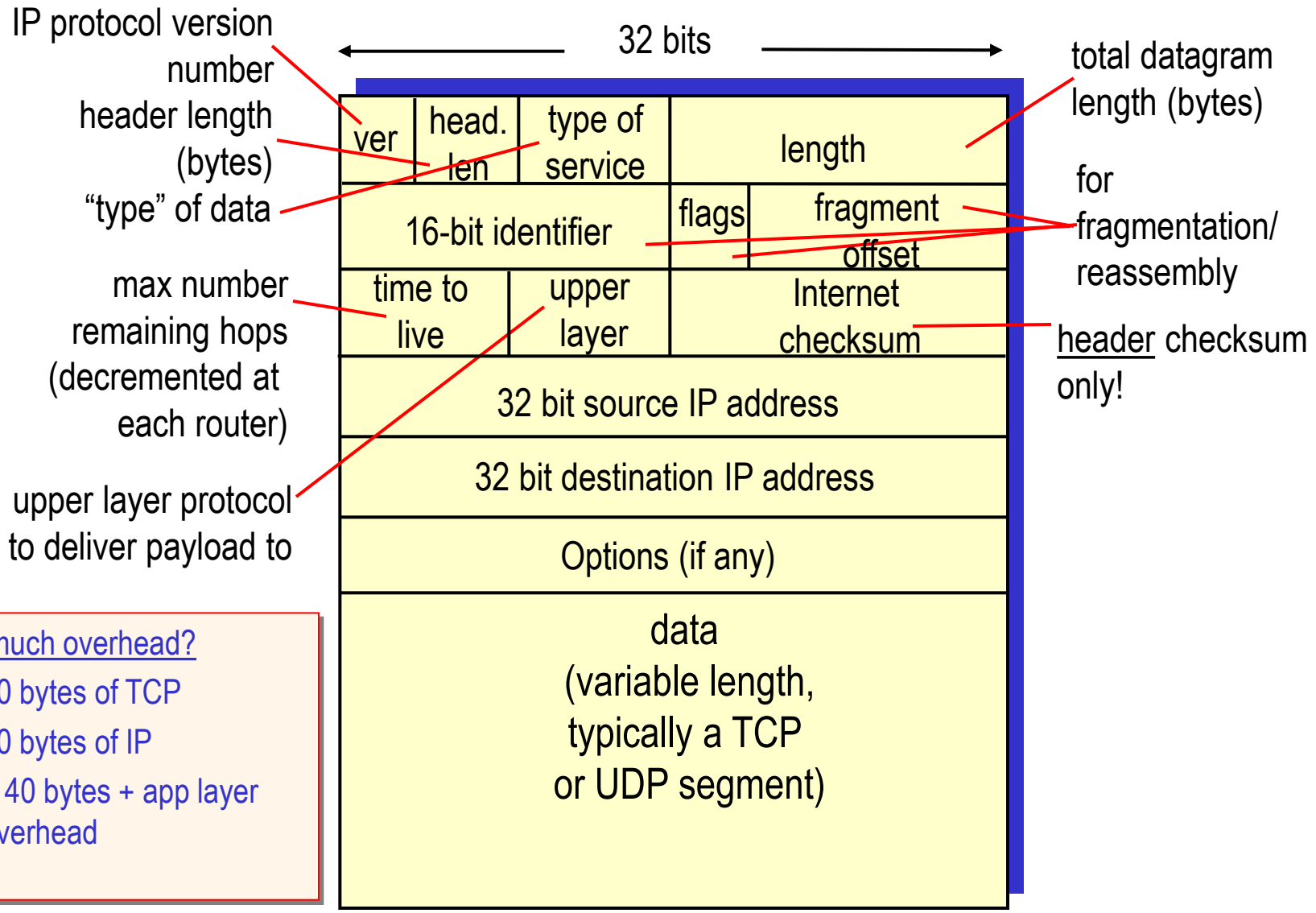
IP addressing: the last word...

Q: How does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers

- allocates addresses
- manages DNS
- assigns domain names, resolves disputes

IP datagram format



How much overhead?

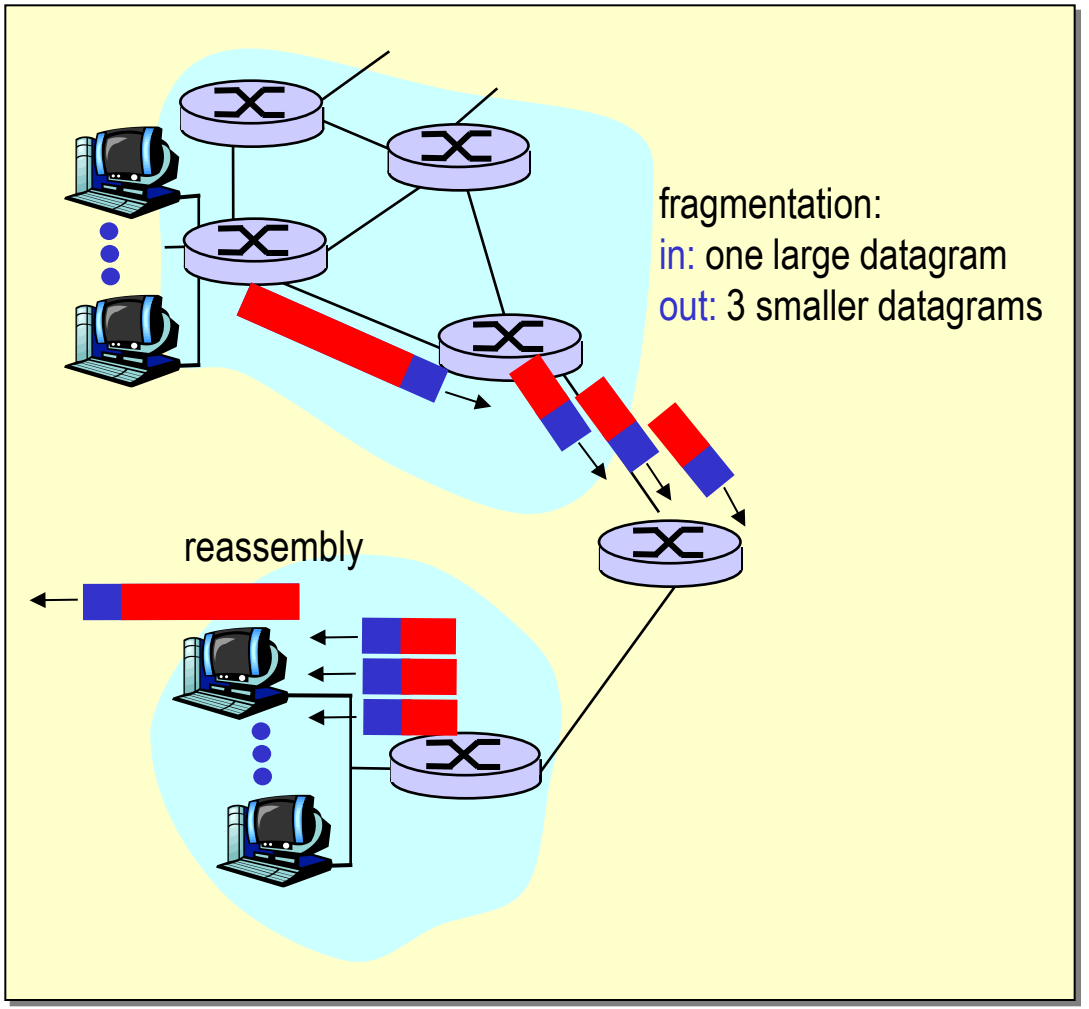
- 20 bytes of TCP
- 20 bytes of IP
- = 40 bytes + app layer overhead

IP header checksum

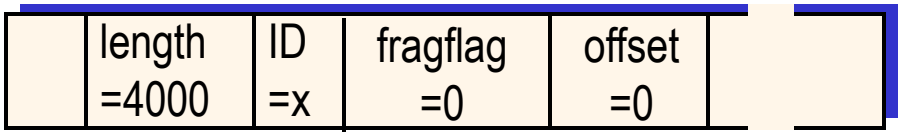
- *Before transmitting the packet, the sender computes a 16-bit checksum of the bits in the IP header and includes the result in the checksum field.*
- *Routers alone the path update the header checksum, to account for any changes in the header, such as decrementing the TTL field.*
- An IP packet with incorrect header checksum is discarded by the receiving machine/router.
- The IP checksum applies only to IP headers and not to the contents of the packets.
- Checking for corrupted data can be provided by the higher-level protocol, such as UDP or TCP, with a separate checksum.

IP Fragmentation & Reassembly

- network links have MTU (max.transfer size) - largest possible link-level frame.
  - different link types, different MTUs
- large IP datagram divided ("fragmented") within net
  - one datagram becomes several datagrams
  - "reassembled" only at final destination
  - IP header bits used to identify, order related fragments



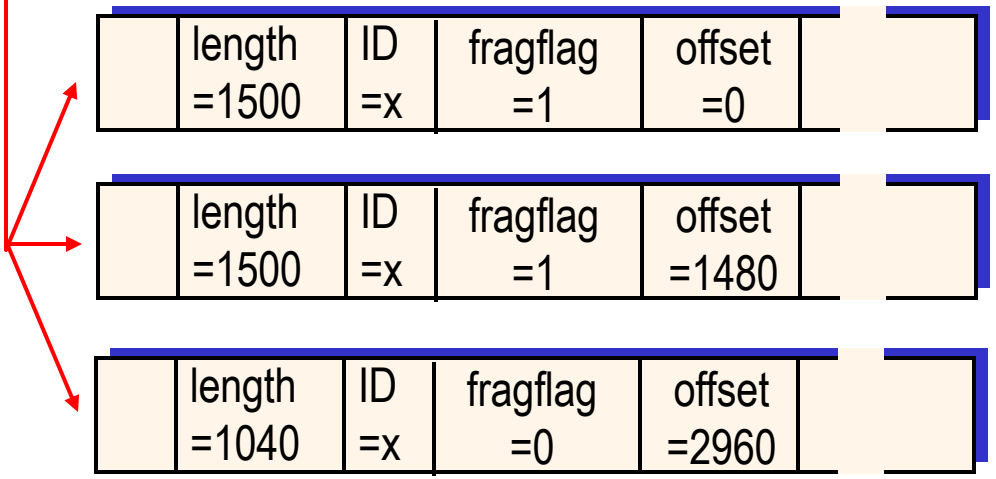
IP Fragmentation and Reassembly



Example

- 4000 byte datagram
- MTU = 1500 bytes

One large datagram becomes several smaller datagrams



Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

במקרים מסוימים poisoned reverse יכול לשפר את הביצועים של אלגוריתם ניתוב מסוג link-state .

1. נכון
2. לא נכון

In some cases, ‘poisoned reverse’ may improve the performance of a link-state routing algorithm.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

כאשר חבילת IPv4- (datagram) מגיעה לנתב (router), הנתב צריך לחשב מחדש את ה- checksum .

1. נכון
2. לא נכון

When IPv4 datagram reaches a router, the router needs to calculate again the checksum.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

נתבים נותנים בדרך כלל עדיפות לחבילות UDP לעומת חבילות TCP.

1. נכון
2. לא נכון

Routers usually give higher priority to UDP packets over TCP packets.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

נתב יעביר חבילה לכוון היעד על ידי שימוש בכניסה הראשונה בטבלת הניתוב המתאימה לכתובת היעד של החבילה.

1. נכון
2. לא נכון

A router will forward a packet towards a destination using the first routing table entry that matches the packet's destination address.

1. True
2. False

Question from an exam
Circle the correct answer

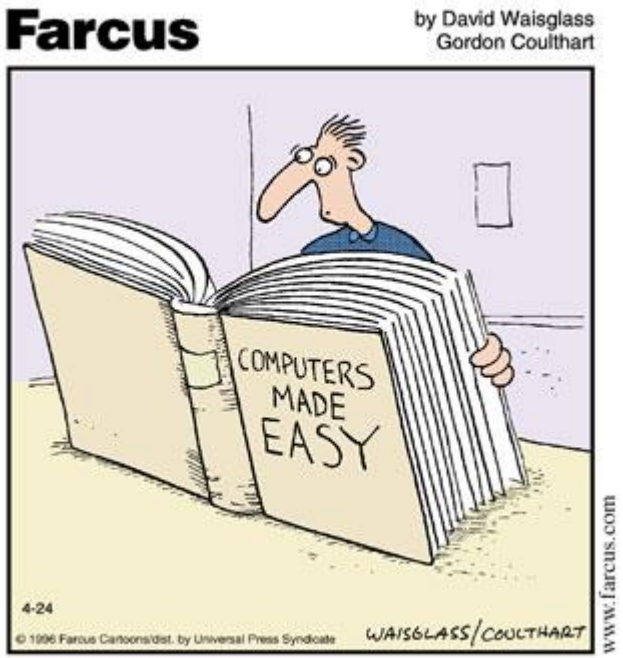
שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

לכל נתב ברשת יש בדיוק כתובת IP אחת.

1. נכון
2. לא נכון

Each router in the network has exactly one IP address.

1. True
2. False


Farcus
by David Waisglass
Gordon Coulthart
COMPUTERS
MADE
EASY
4-24
© 1996 Farcus Cartoons/dist. by Universal Press Syndicate
WAISGLASS/COULTHART
www.farcus.com